CHRONOGYM

Table of Contents

Systems Request Document 3
Requirements Definition Document 4
Use Case 9
DFD 11
Context Diagram 12
ERD 13
Interface Structure Design Diagram 14
Storyboard: Whole System 15

Systems Request Document	
Project Sponsor:	Mark Nguyen
Business Need:	The proposed application is a universal gym crowd tracker. This application allows you to track how busy or crowded the gyms nearby the user's area are. They can also input gyms that they currently go to or look up gyms that they're interested in. The application will consist of statistics that can be inputted by all users for live time updates, and past & current data from the gyms themselves.
Business Requirements:	**Business Requirements:** • Login - Will require the user to make an account (Username, Password, Some sort of authenticator) - Users will set their preferences based on their schedules, location, and gym preferences - Terms and services agreement after signing up • Homepage - Home page will show recommendations based on the user's profile and preferences • Search Engine - Will have extra filters for the users if they have more specific needs - Users will be able to look up specific gyms if needed • Crowd data from gyms - This application will require data from all gyms that want to comply with sharing their crowd data with us
Business Value:	The purpose of this application is to help the users find gyms that are suitable for their schedules or timed workout routines. It could also help gyms bring in more people or balance the crowd out by letting other people discover other locations.

Special Issues:	• Compliance with data from gyms
	  - This application needs many gyms to comply with sharing their crowd data (past, current, and real time)
	• Data integrity on our end
	  - This application needs to keep all data from gyms private and only use it for the sole purpose of the app's function
	  - Needs to keep user data private and protected as well from cyber breaches
	• Real time data
	  - This application may have trouble getting real time data because all gyms use different tracking systems
	• Integrating with different gym systems
	  - The application needs to be able to adapt to the different info systems of the gyms

Requirements Definition Document

Functional Requirements
1. Creating User Login 1.1. There will be an option at the bottom login section that allows the user click on it to "sign up" 1.2. System will require the user's first name, last name, email, and a unique password 1.2.1 The system will store the information into the User database as follows: FirstN, LastN, Email, Password, Username 1.2.2 The password must have at least 1 capitalized letter and 1 special character 1.2.2.1 If the password does not contain either one, an error message will pop up saying they're missing one the requirements 1.3. System will check if there is a duplicated email address in the User database 1.3.1 If the user input a duplicated email, it will send them an error and asks them to login to an existing account or input a new email 1.4. After the user clicks "Finish signing up" and confirms their information, a confirmation email will be sent with a verification code to verify their account 1.4.1 VCodes DB links with User DB 1.5. Once the user confirms it, their new account will be stored into the User database 1.6. The user will receive a confirmation email saying they have successfully signed up and a follow up "Thank you" message

2. Validate User Login

2.1. System will require the user to input their email as their username and password into the appropriate fields

2.2. System will check if the inputted email and password matches with an account in the User database

2.2.1 If succeeded, the system will email them a randomly generated 6 digit verification code for security purposes

2.2.1.1 If the user succeeds in verifying, the system will log the user in and sends them to the homepage

2.2.1.2 If the user fails in verifying, the system will print out an error saying wrong code

2.2.1.3 There will be a "Resend new code" option in case the verification code from the first time didn't go through

2.2.1.4 All verification code related actions are linked between User DB and VCodes

2.2.2 If error occurs such as incorrect username and/or password inputted, it will shoot out an error message indicating the specific error

2.3. There will be a "Forget Password" button that the user can click on to reset their password

2.3.1 The user will need to input an email address that's registered in the User database to receive the email to reset their password

2.3.1.1 If the email address is not in the User database, user won't be able to receive an email and an error message would appear

2.3.2 The password will also be updated in the User database

3. User Account Preferences

3.1 System will display the user's information that is saved in the User database

3.1.1 Users can update their information as well as change their password without verification

3.1.1.1 The password must be unique with at least 1 capitalized letter and 1 special character

3.1.1.1.1 If the password does not contain both requirements, an error message will pop up and the current password will remain the same

3.2 User will have a preferences section underneath their main profile information

3.2.1 User will have the option to set their location or use their current live location

3.2.1.1 User location data will be stored in the UserLoc Database

3.2.2 User can also set preferences to courts or equipment which is linked with CourtsOffered DB and Equipment_Availability DB

3.3 These preferences will be used by the system when the user searches up nearby gyms or filtering results

4. Homepage Layout

4.1 System will display a search bar at the top of the page

4.2 System display nearby gym recommendations to save based on the preferences they set on their profiles below the search bar

4.3 Recommendations will be generated using data stored in the Preferences Database

4.4 System will display a setting icon at the top right of the page with the following options:

4.4.1 User Information & Preferences which directs them to their profile and preferences settings

4.4.2 Sign out option to allow the user to login, and directs them to the login page

5. Saved Gyms Page
   - 5.1 System will display the nearby gyms they have saved through their searches below the recommended gyms
     - 5.1.1 System will store saved gyms through the SavedGyms database where it is linked with UserID and GymID
     - 5.1.2 System will allow users to remove saved gyms from their list anytime they want
   - 5.2 System will display the live time capacity of the gyms saved by the user
     - 5.2.1 The live time capacity will display 3 types of different levels which are “Not Busy”, “Moderately Busy”, and “Very Busy”
     - 5.2.2 System will retrieve real time data from the GymCapacity database where GymID is linked
       - 5.2.2.1 GymCapLevel DB and UserReport DB are also linked with GymCapacity to provide input for real time gym capacities
     - 5.2.3 If a gym does not or can’t provide real time data, the system will display “N/A”
   - 5.3 System will allow users to click on a saved gym to view additional information such as hours, amenities, and past crowd trends UR

6. Gym Search
   - 6.1 System will allow users to search for gyms using keywords or aspects of them such as location, equipment, courts offered, crowd levels, etc.
     - 6.1.1 There will be a Search DB that will store keywords and other search preferences which are linked
     - 6.1.2 This data can be searched through the Gym database from Address, Equipment, CourtsOffered
     - 6.1.3 All of these are linked with GymAddress DB, CourtsOffered DB, Equipment_Availablity DB which are connected to Gyms DB
     - 6.1.4 Crowd levels can be searched in the GymCapacity DB which is linked with Gym_ID
   - 6.2 System will retrieve relevant gyms from the Gym database based on user input
   - 6.3 System will allow users to filter search results based on preferences such as gym type, amenities, and live crowd data
     - 6.3.1 All of these are linked with CourtsOffered DB, Equipment_Availablity DB which are connected to Gyms DB
   - 6.4 System will display search results with gym name, address, current crowd status, and the gym’s distance from the user
     - 6.4.1 This data will be retrieved from the Gyms database in the order of Name, Address, Current_Capacity, Distance

7. Real-Time Data Integration
   - 7.1 System will collect real time crowd data from gyms that agree to share the information
     - 7.1.1 Real time data will be stored in GC_GymCapacity_Input
   - 7.2 System will store the gym’s past and current crowd data in GymCapacity database that are linked to GymID
   - 7.3 If a gym does not provide or cannot provide real time data in the moment, the system will allow users to contribute to the live capacity updates
     - 7.3.1 User contributed data will be stored in the UserReport database with UserID, GC_ID, UR_Capacity Input
     - 7.3.2 System will go through multiple reports before updating the live capacity status
     - 7.4 Capacity will be displayed

8. User Ratings
   8.1 System will allow users to leave reviews and rate gyms out of 5 stars that they have visited
      8.1.1 Ratings will be stored in the Ratings DB which is linked with GymReviews DB
   8.2 System will store user reviews in the GymReviews database which are linked to UserID and GymID
   8.3 System will allow users to edit or delete their own reviews
   8.4 System will display average ratings and the posted reviews when the user clicks on the gym's page
      8.4.1 System will store ratings in the Ratings database which is linked to UserID and GymID
   8.5 System will limit users to one review per gym they had visited
   8.6 System will store reviews and ratings in the Reviews database which is linked to UserID and RatingsID

9. Gym Profiles 9.1 System will display detailed gym profiles when a user clicks on a gym they're interested in
   9.2 Gym profiles will include gym name, address, available amenities, live capacity data, and user ratings
   9.3 System will retrieve this data from the Gyms, GymCapacity, GymReviews, Ratings, GymAddress, CourtsOffered, and Equipment_Availability databases
   9.4 If real-time crowd data is unavailable, system will display a "N/A"

Use Case

Use case Name:_Homepage Layout_____ **ID**_____**4**____ **Priority Level**__**Medium**_____

Primary Actor: _User_____

Process Description:

Homepage will display once the user logins into their account

Trigger: User logs intro their account

Preconditions:

1. User inputs their login information
2. System logs them in
3. The system automatically sends them to the homepage

Normal Course:

1. System will display a noticeable search bar at the top of the homepage
2. System will display recommendations of gyms nearby based on their user settings and preferences below the search bar
3. Recommendations will also display real time capacity levels
4. System will be updating recommendations based on the capacity levels from less crowded to most crowded
   System will also be updating recommendations based on the user's current location
5. System will display a settings icon on the top right displaying the following clickable options of going into their user settings and signing out of their account

Information for Steps

- System Display
- Preferences DB
- Gym Capacity DB
- Update Capacity DB
- User DB
- Update User DB

Alternative Course:

N/A

Use Case

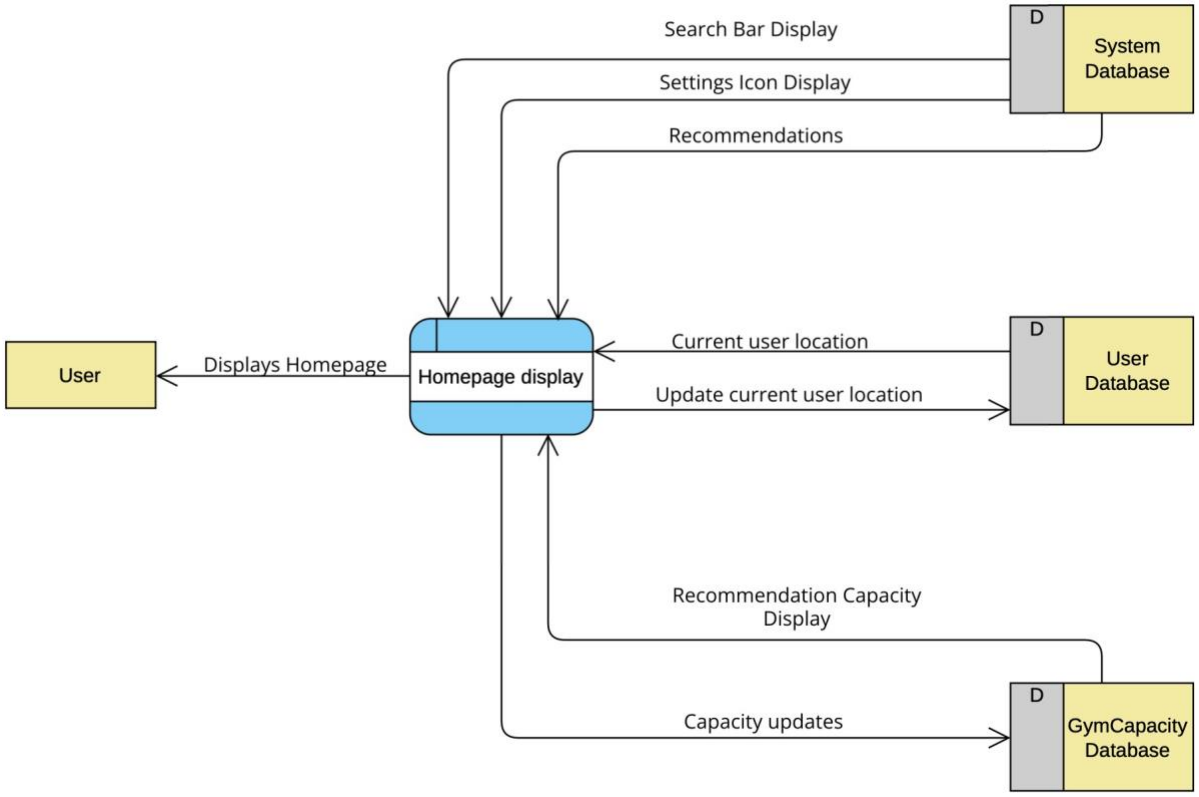
Post Conditions: User will be able to access any elements on the homepage

Exceptions: (if applicable) N/A

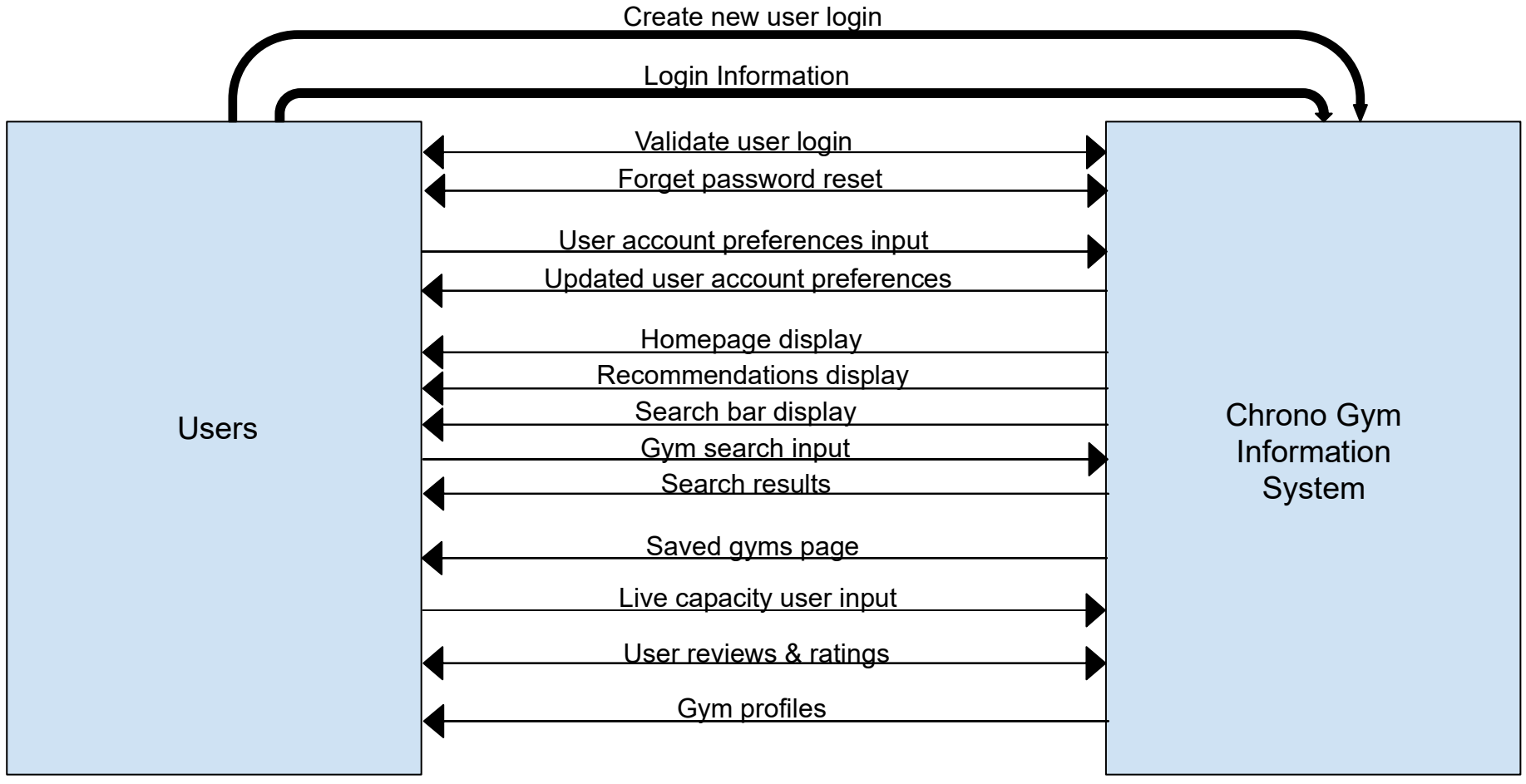
Summary Input	Source	Outputs	Destination
Homepage layout	System	Capacity Updates	GymCapacity DB
Recommendations	System, GymCapacity DB	User current Location	User DB
Capacity levels	GymCapacity DB		
Settings Icon	System		

Use Case

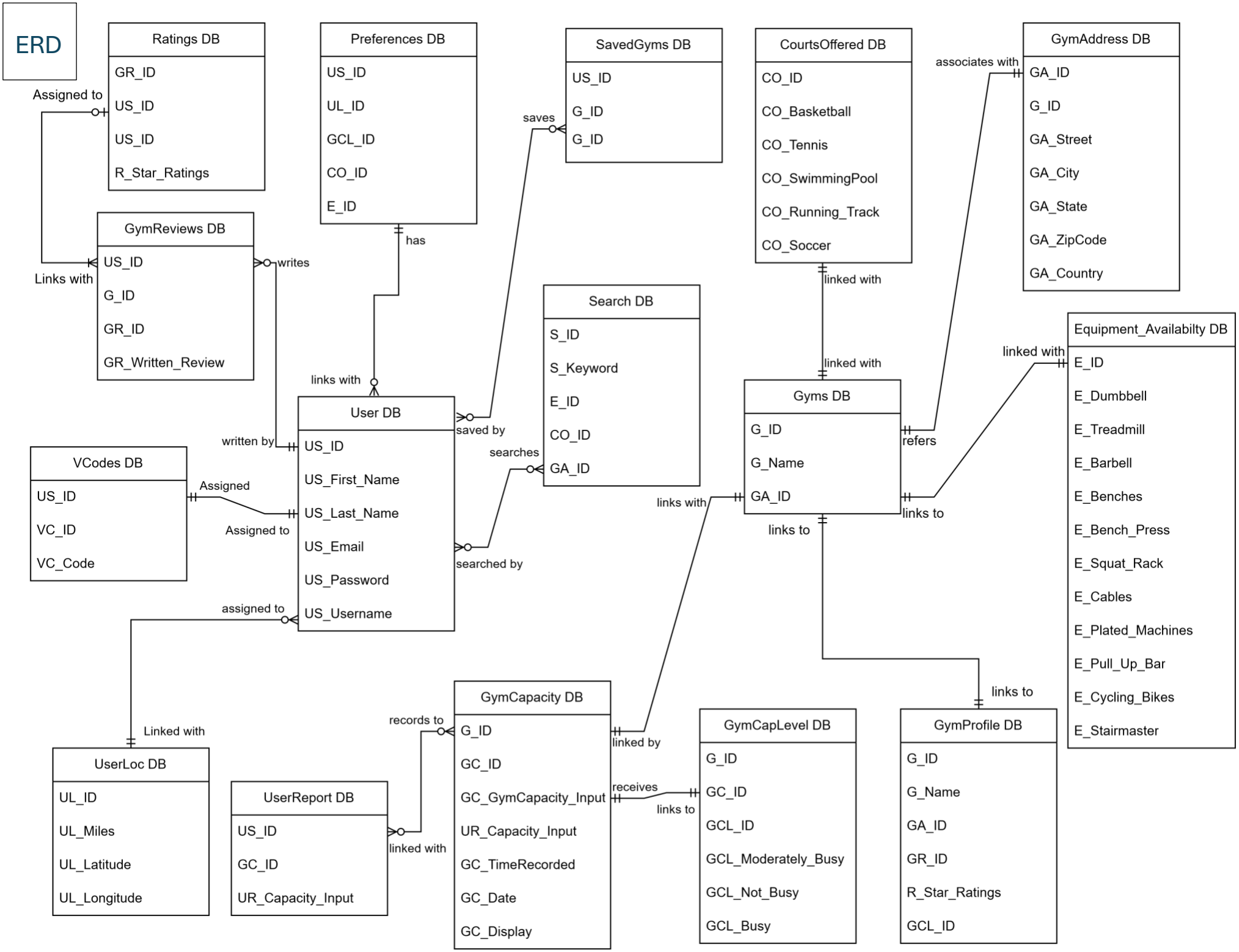
DFD


Search Bar Display
Settings Icon Display
Recommendations
D
System Database
User
Displays Homepage
Homepage display
Current user location
Update current user location
D
User Database
Recommendation Capacity Display
Capacity updates
D
GymCapacity Database

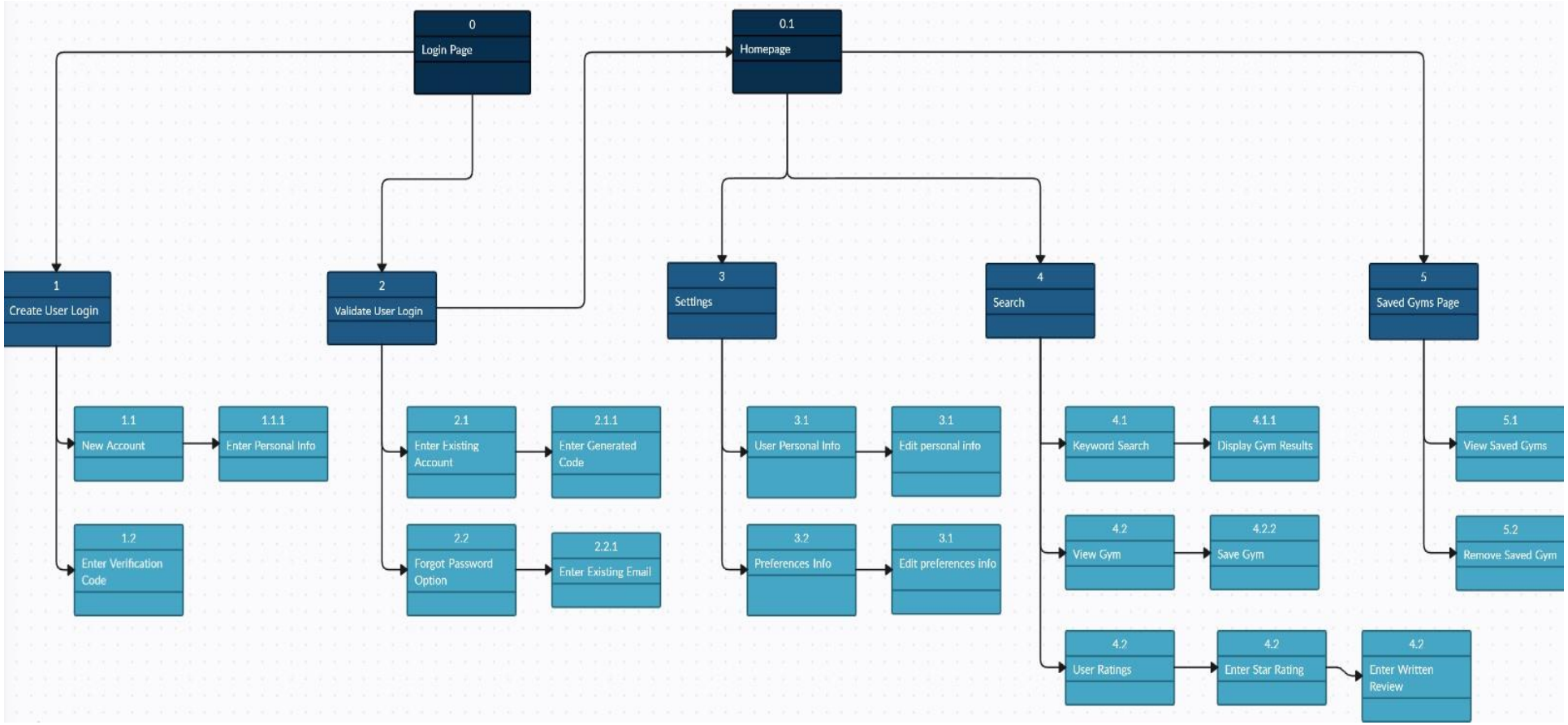
Context Diagram



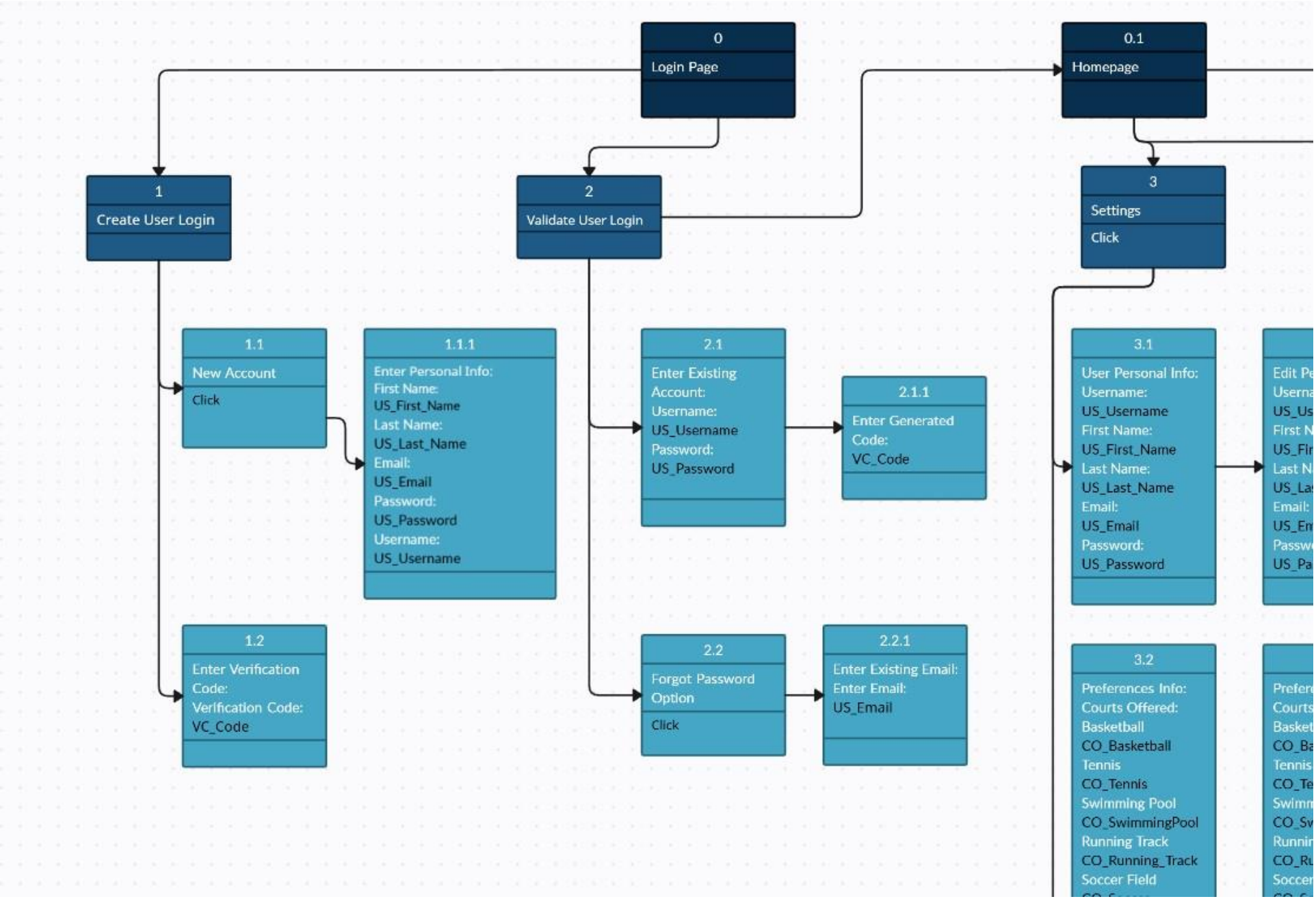
ERD

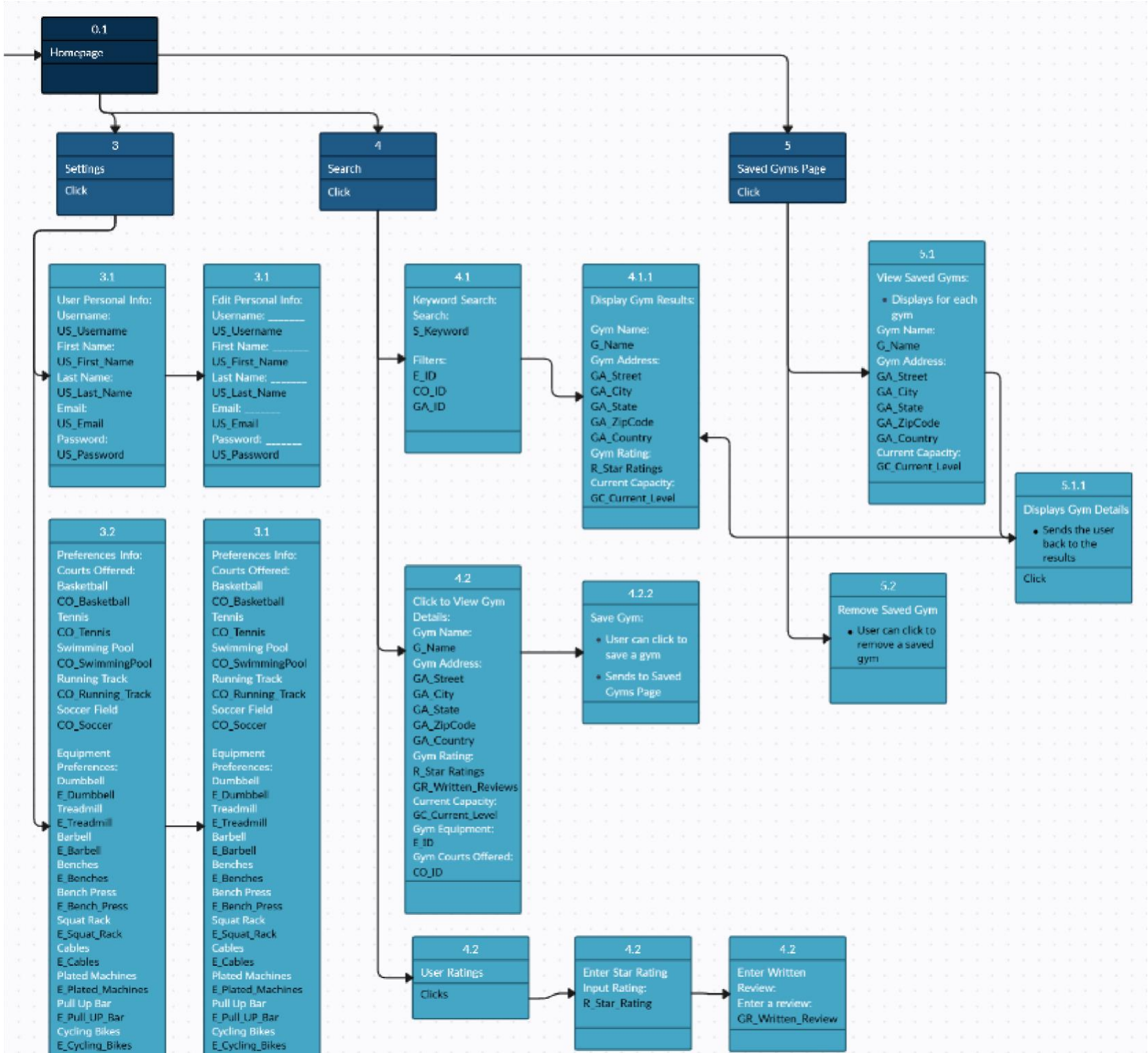

Ratings DB
GR_ID
US_ID
US_ID
R_Star_Ratings
Assigned to
Preferences DB
US_ID
UL_ID
GCL_ID
CO_ID
E_ID
has
SavedGyms DB
US_ID
G_ID
G_ID
saves
CourtsOffered DB
CO_ID
CO_Basketball
CO_Tennis
CO_SwimmingPool
CO_Running_Track
CO_Soccer
associates with
GymAddress DB
GA_ID
G_ID
GA_Street
GA_City
GA_State
GA_ZipCode
GA_Country
GymReviews DB
US_ID
G_ID
GR_ID
GR_Written_Review
Links with
writes
linked with
linked with
Search DB
S_ID
S_Keyword
E_ID
CO_ID
GA_ID
Equipment_Availabilty DB
E_ID
E_Dumbbell
E_Treadmill
E_Barbell
E_Benches
E_Bench_Press
E_Squat_Rack
E_Cables
E_Plated_Machines
E_Pull_Up_Bar
E_Cycling_Bikes
E_Stairmaster
linked with
links with
User DB
US_ID
US_First_Name
US_Last_Name
US_Email
US_Password
US_Username
saved by
searches
searched by
written by
Gyms DB
G_ID
G_Name
GA_ID
refers
links to
links with
links to
VCodes DB
US_ID
VC_ID
VC_Code
Assigned
Assigned to
assigned to
Linked with
links to
UserLoc DB
UL_ID
UL_Miles
UL_Latitude
UL_Longitude
UserReport DB
US_ID
GC_ID
UR_Capacity_Input
linked with
records to
GymCapacity DB
G_ID
GC_ID
GC_GymCapacity_Input
UR_Capacity_Input
GC_TimeRecorded
GC_Date
GC_Display
linked by
receives
links to
GymCapLevel DB
G_ID
GC_ID
GCL_ID
GCL_Moderately_Busy
GCL_Not_Busy
GCL_Busy
GymProfile DB
G_ID
G_Name
GA_ID
GR_ID
R_Star_Ratings
GCL_ID

Interface Structure Design Diagram



Storyboard: Whole System




0.1
Homepage
3
Settings
Click
4
Search
Click
5
Saved Gyms Page
Click
3.1
User Personal Info:
Username:
US_Username
First Name:
US_First_Name
Last Name:
US_Last_Name
Email:
US_Email
Password:
US_Password
3.1
Edit Personal Info:
Username: _____
US_Username
First Name: _____
US_First_Name
Last Name: _____
US_Last_Name
Email: _____
US_Email
Password: _____
US_Password
4.1
Keyword Search:
Search:
S_Keyword
Filters:
E_ID
CO_ID
GA_ID
4.1.1
Display Gym Results:
Gym Name:
G_Name
Gym Address:
GA_Street
GA_City
GA_State
GA_ZipCode
GA_Country
Gym Rating:
R_Star Ratings
Current Capacity:
GC_Current_Level
5.1
View Saved Gyms:
• Displays for each gym
Gym Name:
G_Name
Gym Address:
GA_Street
GA_City
GA_State
GA_ZipCode
GA_Country
Current Capacity:
GC_Current_Level
5.1.1
Displays Gym Details
• Sends the user back to the results
Click
3.2
Preferences Info:
Courts Offered:
Basketball
CO_Basketball
Tennis
CO_Tennis
Swimming Pool
CO_SwimmingPool
Running Track
CO_Running_Track
Soccer Field
CO_Soccer
Equipment Preferences:
Dumbbell
E_Dumbbell
Treadmill
E_Treadmill
Barbell
E_Barbell
Benches
E_Benches
Bench Press
E_Bench_Press
Squat Rack
E_Squat_Rack
Cables
E_Cables
Plated Machines
E_Plated_Machines
Pull Up Bar
E_Pull_UP_Bar
Cycling Bikes
E_Cycling_Bikes
3.1
Preferences Info:
Courts Offered:
Basketball
CO_Basketball
Tennis
CO_Tennis
Swimming Pool
CO_SwimmingPool
Running Track
CO_Running_Track
Soccer Field
CO_Soccer
Equipment Preferences:
Dumbbell
E_Dumbbell
Treadmill
E_Treadmill
Barbell
E_Barbell
Benches
E_Benches
Bench Press
E_Bench_Press
Squat Rack
E_Squat_Rack
Cables
E_Cables
Plated Machines
E_Plated_Machines
Pull Up Bar
E_Pull_UP_Bar
Cycling Bikes
E_Cycling_Bikes
4.2
Click to View Gym Details:
Gym Name:
G_Name
Gym Address:
GA_Street
GA_City
GA_State
GA_ZipCode
GA_Country
Gym Rating:
R_Star Ratings
GR_Written_Reviews
Current Capacity:
GC_Current_Level
Gym Equipment:
E_ID
Gym Courts Offered:
CO_ID
4.2.2
Save Gym:
• User can click to save a gym
• Sends to Saved Gyms Page
5.2
Remove Saved Gym
• User can click to remove a saved gym
4.2
User Ratings
Clicks
4.2
Enter Star Rating
Input Rating:
R_Star_Rating
4.2
Enter Written Review:
Enter a review:
GR_Written_Review

Welcome!

Please sign in or create an account

Username

Password

Welcome!

Please sign in or create an account

mak

****

Invalid Username or Password. Please try again.

Sign Up

First Name

Last Name

Email Address

Username

Password

Sign Up

Mark

Nguyen

mnguyen@bu.eud

mak

****

Invalid email address!
Please try again.